

The Laser-based Drone Vision Disruption with a Real-Time Tracking System for Privacy-Preservation

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Abstract

The capabilities of drones are increasing every day, as is the ease with which civilians can buy and fly them. Most drones are equipped with a camera that is used by a point-of-view operator and, at the same time, can be used for image capture. The use of drones creates a threat to privacy whereby anyone who can fly a drone can take pictures without permission. This study aims to create a 2-axis tracker system that can recognize a drone and locate the position of the drone camera so that a laser beam can track and dazzle the drone camera. The depth-sensing camera is used to localize the part of the target corresponding to the drone's camera and is created using the YOLOv5 algorithm as a deep learning detector model. The drone's camera range and position are challenging to detect due to its small size. Our adaptive detection method combines drone detection and drone camera detection. The depth-sensing camera provides input in the form of a three-coordinate axis from the target. If only the drone is detected, a predictive algorithm can determine the camera's position for illumination with the laser. Alternatively, if the drone camera is detected, the laser can follow the target's movement more quickly. In this study, a green (520 nm) laser module with adjustable power is used to investigate factors that affect the dazzling range. The computer vision detection algorithm can detect and localize the position of the drone camera up to 500 cm with a confidence level of more than 65%. If the target is in the center of the field of view, the accuracy of the target position can reach 98%. The tracker can follow the drone's movement from 2 m/s to 4 m/s with a maximum error of 1.9 cm from the center point of the drone camera for close range. For long range, the maximum error is 6.2 cm. A laser power of 23.5 mW at 500 cm distance is found to be sufficient to dazzle and track drone cameras.

Keywords: Drone; YOLOv5, Laser; Tracking; Camera.

1. Introduction

In recent years, there has been exponential growth in the use of drones for numerous applications due to their low cost and navigational agility (Ahmed et al. 2022; Mohsan et al. 2022; Maghazei et al. 2022). The public also has easy access to drones for recreation and entertainment such as videography, racing, etc., as evidenced by an increase in drone sales year on year (Merkert and Bushell 2020; Kapustina et al. 2021). Most drones are

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sold with integrated cameras, and it is not known what the cameras are used for (Altawy and Youssef 2017). This poses a significant threat to privacy and security, regarding the audio-visual information collected from other people without permission (Rao et al. 2016; Wang et al. 2016; Serafinelli 2022). There is an important concern to ensure privacy protection on private property. The misuse of drones by unauthorized operators may threaten the safety of others and can cause accidents, damage to property, crime, and violation of privacy.

There has been a growing need for defensive technologies to detect and disrupt unwarranted drone surveillance in progress. That can be seen through the counter-drone technologies that have emerged worldwide and the advancement of drones as a defence system (Park et al. 2021b; Chamola et al. 2021; Castrillo et al. 2022). The core technologies for drone countermeasures are to detect, identify, and track. There are already several solutions available on the market that can identify and locate drones using radar, radio frequency (RF) signals, a global positioning system (GPS), and computer vision (Huang and Gartner 2018; Ali et al. 2019; Lykou et al. 2020). Radar has a high-precision tracking capability for large metallic objects and potentially long-range detection based on reflection of radio waves, but the high price and difficulty detecting smaller drones that fly relatively close to the ground and are composed of many non-metallic parts make this technology challenging for widespread use for drones (Rohling et al. 2017). RF-based detection generally tracks the control signal between the drone and the controller, while the position of the drone is tracked by the GPS information from the drone. However, this method relies on persistent drone transmissions in order to detect and locate the drone (Nguyen et al. 2017). Computer vision has the capability to detect and track the drone with a convolutional neural network (CNN) and deep learning training models (Tiwari and Dubey 2022; Zhao et al. 2022; Svanström et al. 2022a). Currently, the development of practical machine-learning techniques for detection, classification, and tracking have been done only for drones and not for the camera on the drone. Drone tracking is commonly used in defence systems to identify, track, and destroy rogue drones or systems and interfere with communications between the drone and its operator (Souli et al. 2022). While these heavyweight solutions can be effective at disrupting a drone attack, they may risk civilian casualties when using weapons to destroy the drone, and signal jammers can cause interference with other signals. Therefore, such systems are confined to use by the military or in other authorised places, such as airports, emergency facilities, and government centers.

Our study aims to develop a novel counter-drone system that detects a drone, identifies its camera, and disrupts the camera. As the drone flies close to the target being observed, it is not sufficient for the laser dazzling system to merely identify the general location of the drone body, since the laser may aim at the propellers or battery and miss the camera lens altogether. Therefore, at close distances, the dazzling system must more accurately identify the exact position of the camera lens on the drone in order to blind it. As a result, our counter-drone laser-based dazzling system puts a special emphasis on training its algorithms to identify, locate and track the drone's camera in order to blind it and preserve privacy.

This paper is divided into three sections, focusing on drone and camera detection, the real-time tracking system, and vision disruption based on laser dazzling. For tracking of the drone and especially its camera, we leverage a single moving depth-sensing camera. This approach enables us to detect and classify an object to address the problem of fast-moving drones and to detect small objects like a camera on the drone (Du et al. 2017). The depth-sensing camera is proposed to provide sufficient measurement accuracy for identification ability in the region of interest (Heraud et al. 2020; Maru et al. 2021). Measurements of the 3-dimensional target position and identification of an object as a drone are essential to build an accurate tracking system. Deep learning algorithms for computer vision based on CNNs and YOLOv5 (You Only Look Once) are used to create an object detector model to achieve faster region classification for drone detection with high position accuracy. Data augmentation and the effective detection of drone cameras is presented through the training and testing of a dataset of 4000 drone-camera images. The reliability of the depth-sensing camera as a sensing device with the model is investigated by comparing the sensor measurement of the detection distance with actual measurements, and by assessing the algorithm's performance to identify a target as a drone or a drone integrated with a camera.

The parameters investigated are the maximum distance readable by the sensor to detect drones, the sensor's maximum reading angle, and the accuracy of readings with different detection angles. The tests were carried out using four drones of various sizes. Since commercial drones with cameras have wheelbase sizes of 12 to 68 cm, if the distance is too great (> 500 cm), the camera appears too small for the sensor to detect it. To determine a drone's camera position at close range, the camera position is obtained from the sensor readings, while for a long-range drone, it is obtained from bounding box center detection with a position adjustment technique. An adaptive algorithm is built from the position of the drone according to the sensor, while considering the camera's location.

A dual-axis rotary platform is developed as a real-time drone tracking system integrated with a laser and optical sensors. This tracker is designed with PWM (Pulse Width Modulation)-based servo motor with 50-330 Hz operating frequency, 25 kg/cm stall torque, and 180° running angle. The tracker's yaw and tilt angle movement are controlled using Raspberry Pi as a single-board computer (SBC). We study photonic countermeasures using a laser to dazzle the camera and disrupt the drone's vision. By mounting a laser pointer with the tracking system, we measure the amount of error introduced by the tracking system in terms of deviation of the laser's aim from the center of the drone's camera. Further, we study the effects of laser power on CMOS (Complementary Metal-Oxide Semiconductor) cameras that are typically employed with consumer drones. A class III green laser with 520 nm wavelength and output power of 58.4 mW is used with a consumer-grade drone to understand the effects of a typical laser on a drone. Parameters such as laser irradiance, incidence angle, and lighting conditions were adjusted to find the optimum power to dazzle the drone without damaging the camera itself. Light scattering also contributed to the dazzle effect on the drone camera. This information can contribute to the control algorithm for tracking the drone's camera.

2. Related work

Detection of drones may be via acoustic, visual, radio-frequency signal-based, or radar approaches (Taha and Shoufan 2019), generally with the purpose of defending airspaces from drones. Technologies against drones are signal jamming, catching drones using nets, or in an extreme case, aerial takedown (Rothe et al. 2019; Jeon et al. 2022). To our knowledge, no previous research has addressed the concept of autonomous drone onboard camera tracking based on visual data. In response to the detection of a drone, the countermeasure disrupts visual information. The questions are what computer vision methods can detect drone cameras, control systems for trackers, and whether lasers can dazzle drone cameras.

2.1. Drone detection and classification

Drone detection and classification abilities are highest when a target is visible, and camera sensors offer the advantage of providing verification that the detected object is a drone while also providing additional visual information, such as the model, dimensions, and payload that other drone detection systems cannot offer. Machine learning algorithms based on hand-crafted feature-based methods and deep learning algorithms can detect drones and analyze their images (Unlu et al. 2017). Initially, object detection algorithms were based upon the extraction of features upstream, searching for sophisticated representations of features that captured the essence of images with well-known descriptors such as a histogram of oriented gradients (HOG), local binary pattern (LBP), and Fourier Transform (Wang et al. 2019; Seidaliyeva et al. 2020). In recent years, to improve the accuracy of detection, real-time performance, detector response time, and invariance of geometrical images, deep learning-based Convolutional Neural Networks (CNN) have been used for image classification (Cazzato et al. 2020). A significant increase in accuracy can be achieved with the R-CNN (Region-based CNN) algorithm, which includes a two-stage object detector to locate the possible object region and a network to extract it. Further, the Fast R-CNN provides better accuracy and speed because it will detect the object rather

than the entire image based on the interest pooling layer technique. (He et al. 2017). It has extended to Mask R-CNN with bounding box recognition to do masking and prediction for each object (Ren et al. 2015). Currently, a one-step object detector with a single feature map divided into a grid can be used to improve the computing speed, termed YOLO (Redmon et al. 2016).

Previous researchers have carried out real-time drone detection and classification. Increasing the accuracy of the drone detection algorithm can be done by separating the detection of moving objects from the flying object classification based on CNN, using a larger dataset to train the classifier (Seidaliyeva et al. 2020). The real-life video drone identifier model with CNN can be generated using synthetic drone images to reduce the time required for manual labelling with an accuracy rate of 92.4% and a precision of 88.8% with a real-life dataset (Wisniewski et al. 2022). To achieve fast drone detection and track flying objects in the scene, the 2D depth map and Gaussian wavelet filtering method can be used for adaptive matched filtering (Paredes et al. 2020). Fast object detection is also used to detect the presence of drones. The YOLO detection algorithm was successfully created to track drones with a speed of 1.5 m/s with a performance detection accuracy of 77% (Wyder et al. 2019). The deep learning based on YOLOv3 is associated with hypothesis tracking methodology with simple online and real-time tracking. The Kalman filter can track small multi-rotor drones at a close distance (Hossain and Lee 2019). Another study also shows the YOLO architecture is preferred because it requires fewer memory resources and detects small drones with multiple bounding boxes per grid cell with better accuracy (Liu et al. 2021; Alsanad et al. 2022). With 106 convolution layers and epochs, training stages can provide an accuracy of up to 99% (Madasamy et al. 2021).

2.2. Tracker and control system

Generally, a device to track a moving object needs 2 or 3 degrees of freedom, depending on the application. Tracking by visual image processing usually uses the image data from the camera to classify objects and detect them. The coordinates of the object are needed to follow the movement of the detected object. Laser ranging can be added to determine drone altitude and calculate longitude and latitude (Sheu et al. 2019). In addition, a depth-sensing camera can also be used to determine the drone's position (Svanström et al. 2022b). The tracker typically uses the pan-tilt or gimbal on a drone, controlled by 2 or 3 stepper or servo motors. The focus of research on trackers is on the control algorithm to follow objects. The movement of the detected object can be seen from the deviation of the target from the image center, and this value can be used as a reference point. It can be combined with a Gimbal control algorithm based on motion compensation (Lin et al. 2021). An active disturbance rejection control technique can be combined with a tracking control system to improve tracking ability (Chen et al. 2015). The camera mounted on the tracker must be stable while its line-of-sight points at a target. For this reason, the mechanism's kinematics and imaging geometry should be decoupled (Huynh et al. 2021). The input delay problem can affect trackers for sensing, so the tracker is unable to respond quickly to follow the target. This problem can be overcome with a non-linear backstepping control design and vision-based neuro-fuzzy controller (Qadir et al. 2014; Park et al. 2021a).

2.3. Photonic countermeasures

High-powered laser pointers have the potential to dazzle the color channels of cameras preventing the production of images and even causing complete sensor failure. Using a suitable laser system with appropriate wavelength and power requirements can effectively address privacy concerns while averting unintended damage to the drone or its operator (Benoist and Schleijpen 2014; Ritt and Eberle 2015; Zhang et al. 2015). A coherent light source such as a laser effectively creates a dazzling effect on CMOS imaging devices since they are susceptible to high-intensity light fluxes. Various studies have shown the capability of lasers in both the visible and NIR (Near Infrared) spectrum to distort image quality and, with enough power, damage the optical

sensor of a camera (Nascimento Santos et al. 2018a). Comprehensive testing of the CMOS sensors performed at varying distances in airborne scenarios has successfully dazzled the cameras. By creating variations in flux and light intensity, beam positioning, and irradiance level, the number of dazzled pixels increased throughout the exposure. This resulted in a large portion of the camera's field of view (FoV) being entirely blinded by the laser, to produce the expected dazzle. This demonstrates the effectiveness of photonic countermeasures in an airborne scenario of CMOS cameras commonly found in modern consumer drones (Luo et al. 2019; Eberle et al. 2019). Interference and distortion of the image occur at higher irradiance levels, causing lens flares and scattering effects. Due to the lens flare from the camera, in some cases, the laser did not need to be perfectly aligned to the focal point of the camera, and dazzling occurred from angles extending to only half of the field of view of the camera (Nascimento Santos et al. 2018).

3. Tracking Architecture and System

We first describe the design of our drone tracker architecture and control system. The tracker was designed to be placed stationary on a raised area and to rotate about two axes, i.e., 360° of yaw rotation and $+75^\circ$ of tilt. The limit on the tilting movement is because a drone's camera tends to capture images that are diagonally below the drone, and it is impossible to blind the drone camera if its position is perpendicular to the blinding light source. Fig. 1 shows the design architecture of the tracker with the laser pointed in the same direction as the camera, where there is a distance of D_{LC} on the z-axis between the laser and the depth-sensing camera.

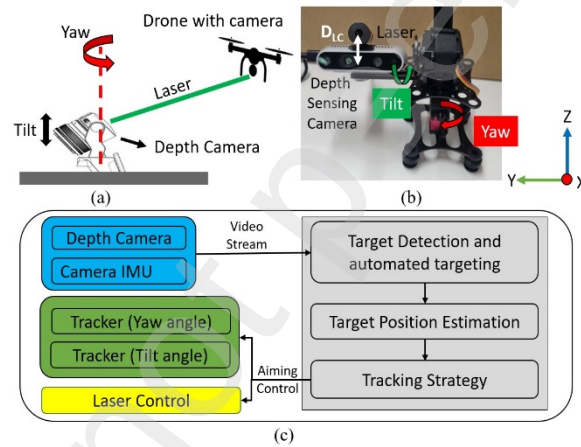


Fig. 1. Drone tracker architecture: (a) Illustration of working principle; (b) Depth-sensing camera and laser position; (c) Block diagram system.

In this design, the sensing device is an Intel RealSense depth-sensing camera D435 connected to a processing computer with a dedicated GPU for fast processing. The results are used to track the drone's movements, and the laser is activated when the drone is detected. In this design, the tracker is controlled by a Raspberry Pi and connected to a servo motor, with cloud-based communication between the microprocessor and the computer.

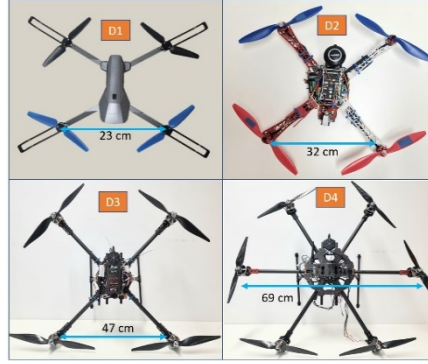


Fig. 2. The drones used in this experiment.

This work focused on multi-object classification to detect a drone and its camera together. The depth-sensing camera detects reflected points from the target in X, Y, and Z coordinates, and a deep-learning algorithm classifies the object. The reliability of vision sensor results based on the deep-learning algorithm was tested by measuring the distance range to detect drones and onboard cameras, and the accuracy of reading the drone's position at different angles. The experiments were carried out using 4 drones of various sizes, as shown in Fig. 2.

3.1. Depth-sensing camera

Vision was selected both to sense the drone and to estimate the drone's distance and camera position. We used an open-source camera with documentation and development environments for the python programming language. The Raspberry Pi Camera Module performs image detection and classification (Miikki et al. 2021). The distance of the detected object is obtained by determining the size of the object through its bounding box relative to the overall size of the frame and translating it into a distance. The precision is less than for a depth-sensing camera which can provide accurate distance estimates via the depth of the detected object (Tadic et al. 2022). The use of two sensors makes it possible to obtain the position of an object in 3 dimensions based on point cloud calculations. The RealSense depth-sensing camera D435 from Intel is used because it has a global shutter that ensures better performance in situations of dim light, while capturing fast movements in a scene and reducing image blurring. The D435 depth-sensing camera shows superior depth measurement results when the targeted objects are located a few meters away from the camera, and it has a broad field of view (FOV): 85° (Horizontal) x 58° (Vertical) x 90° (Diagonal). The accuracy of the measurement is given as the percentage of the error in the depth measurement, while precision is the capability of the sensor to detect the exact size under identical conditions.

Using stereo cameras, two images from the two sensors are compared and can be calculated in x, y, and z coordinates of any reflected point, as illustrated in Fig. 3. The left and right cameras have the exact specifications with reflected angle α , β , and γ in epipolar lane toward the target. The depth information is derived from the comparisons since the distance between the sensors is known ($D_s = 5 \text{ cm}$). The correlation between position and angle is shown in Equation (1), derived from the law of sines (Rodríguez-Quinonez et al. 2017).

$$(1) \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = D_s \begin{bmatrix} \frac{\sin \gamma * \sin \beta}{\sin(\beta + \gamma)} & \frac{1}{2} - \frac{\sin \gamma * \cos \beta}{\sin(\beta + \gamma)} & \frac{\sin \gamma * \sin \beta * \sin \alpha}{\sin(\beta + \gamma)} \end{bmatrix}^T$$

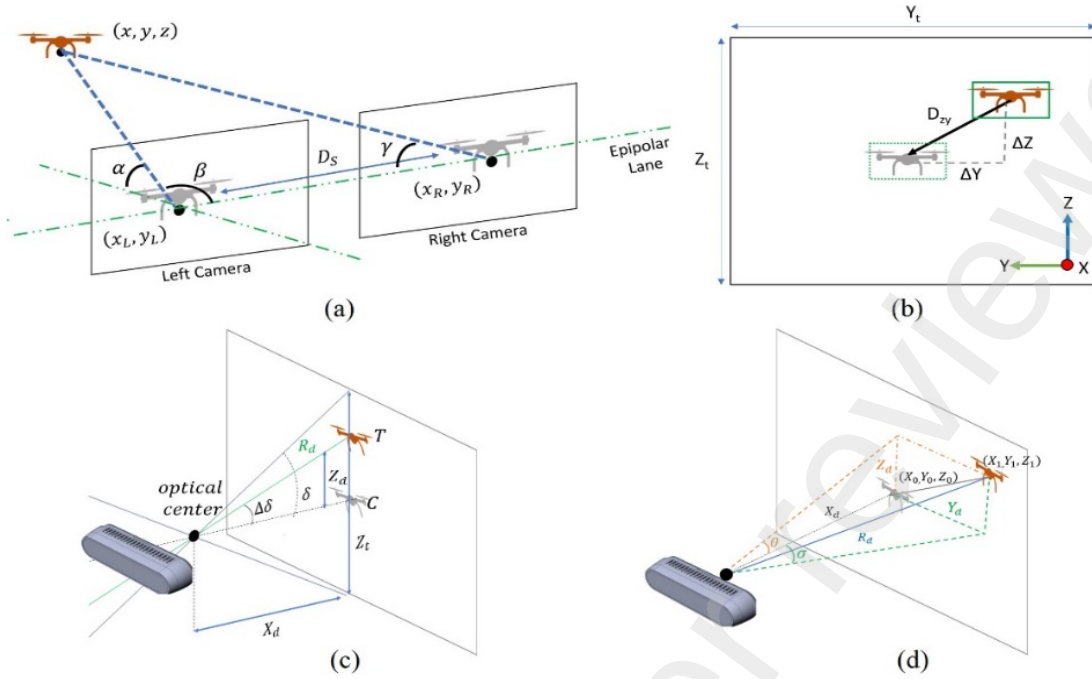


Fig. 3. Illustration of vision sensor: (a) drone detection with depth-sensing camera (b) The displacement between target image center and actual position (c) The relation between angle and target position (d) The angle displacement.

The tracking control model designed continuously positions the drone at the center of FoV by calculating the amount of displacement (D_{zy}) between the actual target position (T) and the image centroid (C), where Y_t represents the FOV on the y-axis, the FOV on the z-axis is represented by Z_t , and X_d is the detection distance between the vision sensor and the target on the center of FoV. From the perspective of the depth-sensing camera, we consider the relationship between the FoV reading angle and the image width given in Equations (2) and (3), as shown in Fig. 3(c) and 3(d), to calculate the displacement. The depth-sensing camera's angle of view (AoV) is represented by δ , and the AoV rotation to detect the actual target is $\Delta\delta$. In Fig. 3(b), horizontal and vertical displacement targets are ΔY and ΔZ , respectively.

$$\frac{Z_t}{2} = kX_d \tan\left(\frac{\delta}{2}\right) \rightarrow Z_t = 2kX_d \tan\left(\frac{\delta}{2}\right) \quad (2)$$

$$kX_d \tan(\Delta\delta) = \Delta Z \quad (3)$$

In some cases, if the distance between the vision sensor and the target cannot be determined, the size of the bounding box is a reference for the distance. The difference in scale between the actual target image and the initial image is represented by k . According to Equations (2) and (3), the magnitude of the movement angle to track the target at the FoV center can be calculated by Equation (4).

$$\Delta\delta = \tan^{-1} \left[\frac{2\Delta Z}{Z_t} \tan\left(\frac{\delta}{2}\right) \right] \quad (4)$$

The tracker movement is based on rotational angles, namely, tilt angle (θ) and yaw angle (σ). Tracker movement angle is controlled using input data of distance and target position, and calculated using Equation (5). The movement illustrations are shown in Fig. 3(d). The positions of the depth-sensing camera and the laser

are separated by D_{LC} , so the angle of vertical movement is adjusted as in Equation (6).

$$\begin{bmatrix} \theta \\ \sigma \end{bmatrix} = \begin{bmatrix} \tan^{-1} \left(\frac{Z_d}{X_d} \right) \\ \tan^{-1} \left(\frac{Y_d}{X_d} \right) \end{bmatrix} = \begin{bmatrix} \tan^{-1} \left(\frac{Z_1 - Z_0}{X_d} \right) \\ \tan^{-1} \left(\frac{Y_1 - Y_2}{X_d} \right) \end{bmatrix} \quad (5)$$

$$\begin{aligned} \begin{bmatrix} \theta \\ \sigma \end{bmatrix} &= \begin{bmatrix} \tan^{-1} \left(\frac{Z_1 - Z_0}{X_d} \right) \\ \tan^{-1} \left(\frac{Y_1 - Y_2}{X_d} \right) \end{bmatrix} + \begin{bmatrix} \theta_{D_{LC}} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \tan^{-1} \left(\frac{Z_1 - Z_0}{X_d} \right) \\ \tan^{-1} \left(\frac{Y_1 - Y_2}{X_d} \right) \end{bmatrix} + \begin{bmatrix} \tan^{-1} \left(\frac{D_{LC}}{X_d} \right) \\ 0 \end{bmatrix} \end{aligned} \quad (6)$$

To measure the accuracy of distance readings from the deep learning model, the target readings were compared when in line with the sensor and with different angles (θ) on the epipolar line, as shown in Fig. 4. Two experiments were conducted: in the horizontal plane where the drone moves left and right and in the vertical plane where the drone moves up and down. The results influence the detection method and the algorithm to follow the drone and disrupt the camera.

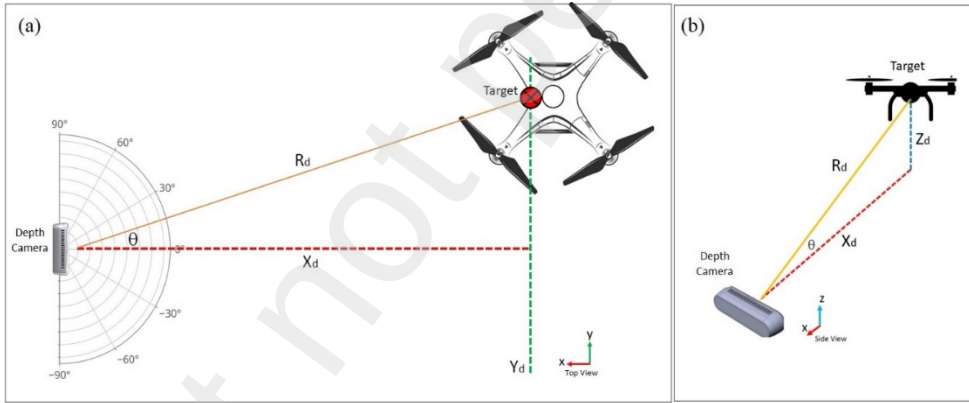


Fig. 4. Illustration of drone detection with a different angle in (a) Horizontal plane (b) Vertical plane.

The two-axis tracker system is mounted on a servo motor with a vision sensor and laser, depicted in Fig. 1(b). The rotation of each motor is controlled according to the flying behavior of the drone. For this work, we focus on tracking drones within the field of view and evaluate the system's ability to track and blind such drones, leaving to future work a general search method to scan the skies for drones. The tracker workspace to control the depth-sensing camera movement can be described with a 2-dimensional kinematic model. This model represents the relation between Z and X positions for tilt (σ) and yaw (θ) angle. The rotational tracker movement in the X and Z -axis is obtained by a kinematic moving frame and can be seen in Equations (7) and (8).

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad (7)$$

$$R_z(\sigma) = \begin{bmatrix} \cos \sigma & \sin \sigma & 0 \\ -\sin \sigma & \cos \sigma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8)$$

The transformation matrices are derived in terms of the total rotation matrix shown in Equation (9).

$$R_T = R_x(\theta) R_z(\sigma) = \begin{bmatrix} \cos \theta & \sin \theta \cos \sigma & \sin \theta \sin \sigma \\ -\sin \theta & \cos \theta \cos \sigma & \cos \theta \sin \sigma \\ 0 & -\sin \sigma & \cos \sigma \end{bmatrix} \quad (9)$$

The equation of motion of the tracker system is calculated based on the Lagrange-Euler algorithm and the kinematic equation for rotational velocity can be written as shown in Equation (10) and (11).

$$\dot{R}_x(\theta) = \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = \begin{bmatrix} X \cos \theta + Y \sin \sigma \\ -X \sin \theta + Y \cos \sigma \\ Z \end{bmatrix} \quad (10)$$

$$\dot{R}_z(\sigma) = \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = \begin{bmatrix} X \\ Y \cos \sigma + Z \sin \sigma \\ -Y \sin \sigma + Z \cos \sigma \end{bmatrix} \quad (11)$$

3.2. Drone camera and laser

The green laser with 58.4 mW output power and 50 kHz modulation frequency is used for the lab experiment with adjustable current starting from 0-200 mA. In the field experiment, the green laser pointer is used as a proxy to shine and track the drone camera, which was replaced for safety reasons. QR (Quick Response) code is used in this study to prove whether the camera can still capture the QR code and how much dazzling effect occurs on the camera. D_{LC} symbolizes the distance between the camera and the laser, and the power of the laser module can be controlled via PGL-F laser module software to control frequency and current. The beam profile has a spatial intensity; therefore, the amount of beam divergence can be calculated using the Gaussian normal distribution. It can be seen in Equation 12.

$$\alpha = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right) \quad (12)$$

The erf is the error function, x is the beam radius, and α is the beam divergence. The beam size depends on the distance between the center of the beam and the center of the drone camera (D_{LC}), the radiuses of the shaft (d_b), and the drone camera aperture (d_t), see Equation. 13.

$$\alpha = \frac{1}{2} \left(\operatorname{erf} \left(\frac{D_{LC} + d_b}{d_t \sqrt{2}} \right) - \operatorname{erf} \left(\frac{D_{LC} - d_b}{d_t \sqrt{2}} \right) \right) \quad (13)$$

We selected several examples of inexpensive consumer drones for our research. Table 1 provides a comparison of the onboard cameras in typical consumer drones, showing the properties of their sensors, lenses, and shutter speeds.

Table 1. Consumer drone camera comparison

Description	Mavic Mini	DJI Mini 2	DJI Spark
Sensor	1/2.3" CMOS	1/2.3" CMOS	1/2.3" CMOS
Effective Pixels	12 MP	12 MP	12 MP
Lens	FOV: 83° 35mm Equivalent: 24 mm Aperture: f/2.8 Shooting Range: 1 m to ∞	FOV: 83° 35mm Equivalent: 24 mm Aperture: f/2.8 Shooting Range: 1 m to ∞	FOV: 81.9° 35mm Equivalent: 25 mm Aperture: f/2.6 Shooting Range: 2 m to ∞
Shutter speed	Electronic shutter: 4 – 1/8000s	Electronic shutter: 4 – 1/8000s	Electronic shutter: 2 – 1/8000s

Most drones use CMOS sensors to produce images for navigation and to capture videos or pictures. Auto-adjustment settings for commercial CMOS cameras adapt to changes in lighting conditions, but this interferes with our experimental scheme. For preliminary experiments, we used a Sony IMX477-AACK-C 12.3 Mega-Pixel CMOS camera with a color image sensor integrated with a Raspberry Pi, and the camera parameters were regulated by the Pi. Typically, cameras are supplied with attached filters, lenses, rolling shutters, and software-corrected auto-adjustments, which should be considered when experimenting with the cameras. The camera we used, shown in Fig. 5, has adjustable image settings (lenses can be added, modified or removed).

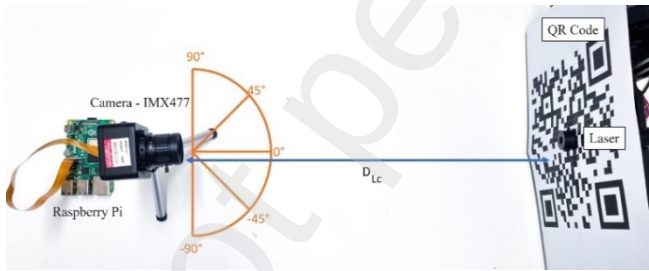


Fig. 5. Laser experiment to dazzle the camera.

In the last stage, the laser is placed on the tracker and integrated with the depth-sensing camera to build a tracker movement control system. By recording the test with the drone camera, the dazzling that occurred can be seen, and the system's reliability to follow the drone's movements and interfere with its vision can be inspected.

4. Drone detection algorithm and vision

4.1. YOLOv5 model

In this study, the embedded system based on YOLOv5 deep learning model is proposed to identify the drone and drone camera with a real-time system. The challenge is to detect a small object, i.e., a drone camera, in real-time. Due to its vast network framework and volumes of data, YOLOv5 exhibits very good performance in image identification. The YOLOv5 network consists of a pre-trained convolutional neural network (CNN), neck, and head. CNN extracts feature from the input image known as the backbone and passes them to the

"neck." In this stage, the image features are mixed and combined before passing them on to the prediction layers to predict targets on the features map.

YOLOv5 has five architecture models, namely, YOLOv5n (nano), YOLOv5s (small), YOLOv5m (medium), YOLOv5l (large), and YOLOv5x (extra-large). The differences between these models are the number of feature-extraction modules, kernels, and neural network parameters that affect accuracy and processing time. Details can be seen in the YOLOv5 repository. In our drone camera detection model, YOLOv5l architecture is used for large models with 46.5 million parameters, an accuracy of 67.3, and a 10.1 ms GPU time. The YOLOv5 networks can detect objects using $S \times S$ grid cells, with $S = 7$. The grid determines whether an object is detected when the target's ground truth center is in the grid. Each grid contains conditional probabilities and a bounding box (with confidence scores). To achieve excellent prediction, YOLOv5 multiplies the individual box confidence precisions with conditional class probabilities, eliminating the low probability bounding boxes and keeping the highest chance in the predicted bounding box. The intersection over union (IoU) unit represents the overlap between the ground truth box (T) with the predicted bounding box (P). The IoU determines the bounding box's contents and calculates the target confidence value. The confidence value (σ) of the bounding box is calculated in the center of the bounding chest (T_0) as shown in Equation (14).

$$\sigma_{T_0} = P_r(\text{Target}) \times IoU_P^T; P_r(\text{Target}) \in (0, 1) \quad (14)$$

The predicted target (P_r) value is represented by 0 or 1 to show whether there is a target object in the bounding box. The confidence value lies between 0 and 1; the closer it is to 1, the higher the accuracy of the object detection. The uniqueness of YOLO is its ability to multiply individual bounding boxes based on conditional class probabilities to get accurate predictions.

$$\begin{bmatrix} B_x \\ B_y \end{bmatrix} = \begin{bmatrix} \zeta(T_x) + C_x \\ \zeta(T_x) + C_y \end{bmatrix} \quad (15)$$

$$\begin{bmatrix} B_W \\ B_H \end{bmatrix} = \begin{bmatrix} P_W e^{T_W} \\ P_H e^{T_H} \end{bmatrix} \quad (16)$$

The prediction of an object is followed by the predetermined ratio of bounding boxes and then by the coordinates of each bounding box in the output feature map, i.e., box width (B_W), box height (B_H), center of the bounding box ($B_x; B_y$), and the object class (C). The training process with the CNN generates the ground truth (T_*) value to predict the coordinates for each bounding box, with offset values at the top left corner of the target ($C_x; C_y$), prior width (P_w) and height (P_h) of the bounding box. These values are calculated in the sigmoid activation function (ζ_*) or exponential function (e^*) from Equations (15) and (16).

4.2. Detection of the drone and drone camera

A pre-trained YOLOv5 model provides drone and camera datasets for training with 4000 drone images, 106 convolution layers, and 45,000 epochs to support high-accuracy detection. The supervised learning method annotates the drone and drone camera images as a correct target output. This step is performed by manual labeling from 4000 image data collection, and the YOLOv5 optimization algorithm model will learn from the labeled data by identifying patterns to do model training. Fig. 6 shows examples of the performance accuracy of target classification in both stall and flying states at 400 cm distance. YOLOv5 provides bounding boxes of the estimated locations of drones in the image in Figure 6a, and of both the drone and its drone camera in Figures 6b and c. Also, the confidence of the estimate is given next to the respective bounding boxes. Our model, used

with a depth-sensing camera, achieved an accuracy rate of more than 80% for detecting the drone and a confidence rate of more than 0.7 for the camera.

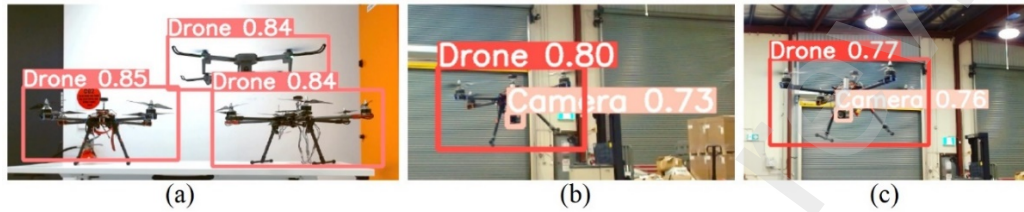


Fig. 6. Drone and camera detection results on the target dataset using the proposed algorithm: (a) Drone detection (b) Drone camera detection (Sample 1) (c) Drone camera detection (Sample 2).

The performance of the object detection models from the drone camera training can be seen through the loss and mean average precision (mAP) and are shown in Fig. 7. The percentage error between the predicted output of the model and the ground truth is represented through the bounding box loss, objectness loss, and classification loss. Our model has less than 0.025 box loss, the confidence loss of object presence (objectness loss), is less than 0.01, and 0.001 classification loss during the training (known as cross-entropy). The results also show the true positive detection performance approaches 0.9, followed by the recall value. The average mAP in the Intersection over Union (IoU) threshold from 0.5 to 0.95 is 0.58.

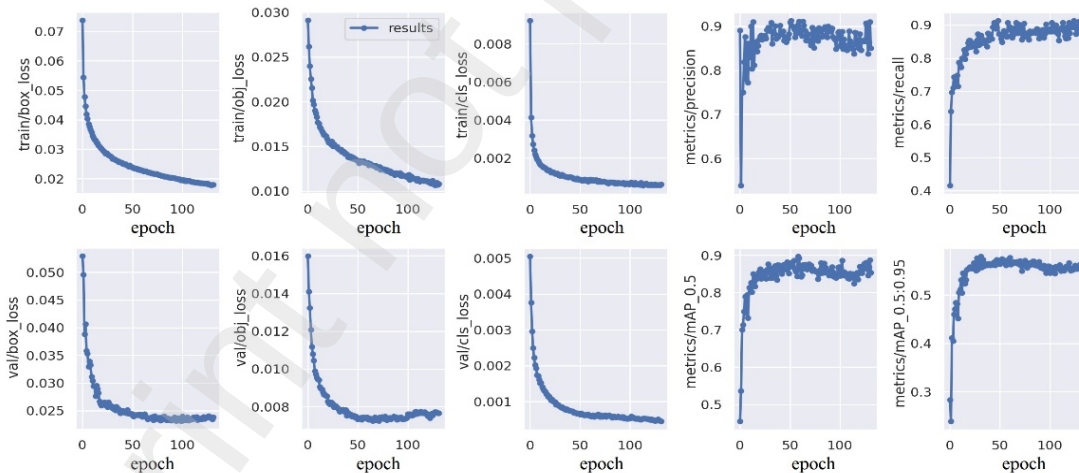


Fig. 7. Feature extraction of drone and camera training.

4.3. Target and distance detection in Convolutional Neural Network-based model with depth-sensing camera

Using the above trained algorithm, we tested the confidence of recognition of the drone and its camera at different distances. The ability to read target coordinates (X-, Y-, and Z-axis) depends on the sensor's reliability and the detection object's size. The test was carried out using four drones of different sizes, as illustrated in Fig. 2. The symbol D1 represents the smallest drone, and the number increased in size, with D4 representing the largest drone used in this study. The test is carried out every 100 cm until the depth-sensing camera cannot detect the drone. The results are shown in Fig. 8. The maximum detection distance for the drones are as follows: D1 is 800 cm, D2 is 1000 cm, D3 is 1945 cm, and D4 is 2120 cm. The detection model designed with the D435 depth-sensing camera can detect drones with a maximum size of 69 cm with a confidence of more than 50% at a maximum distance of 500 cm. The drone camera detected within this distance has more than a 63% confidence level. Since all sizes of drones can be detected within 800 cm, tracker modeling has proven effective at this limit.

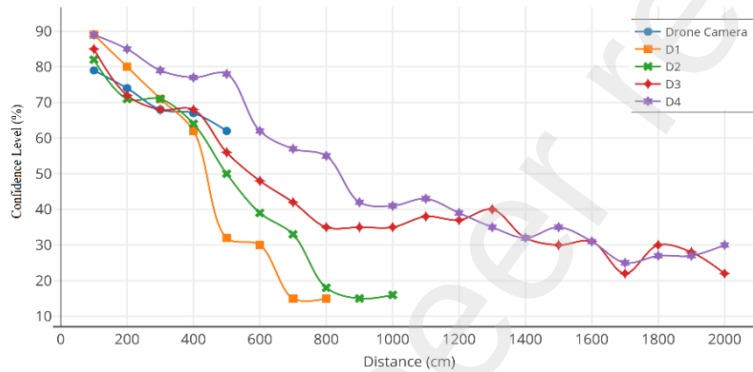


Fig. 8. Target detection and confidence results.

The small size of the drone camera poses a challenge for the YOLOv5 algorithm in terms of detection range. The drone camera was detected up to a maximum distance of 500 cm in the experiments. The reliability of the depth-sensing camera is examined to compare the accuracy of target readings when in line with the sensor and at different reading angles. Fig. 9 showed the test carried out by reading the detection distance between the drone and the sensor with different viewing angles. The horizontal plane is represented by the symbol "H" and vertical plane is represented by the symbol "V". This trial was carried out with four drones with X_d distances of 400 cm and 800 cm, and the results can be seen in Fig. 9.

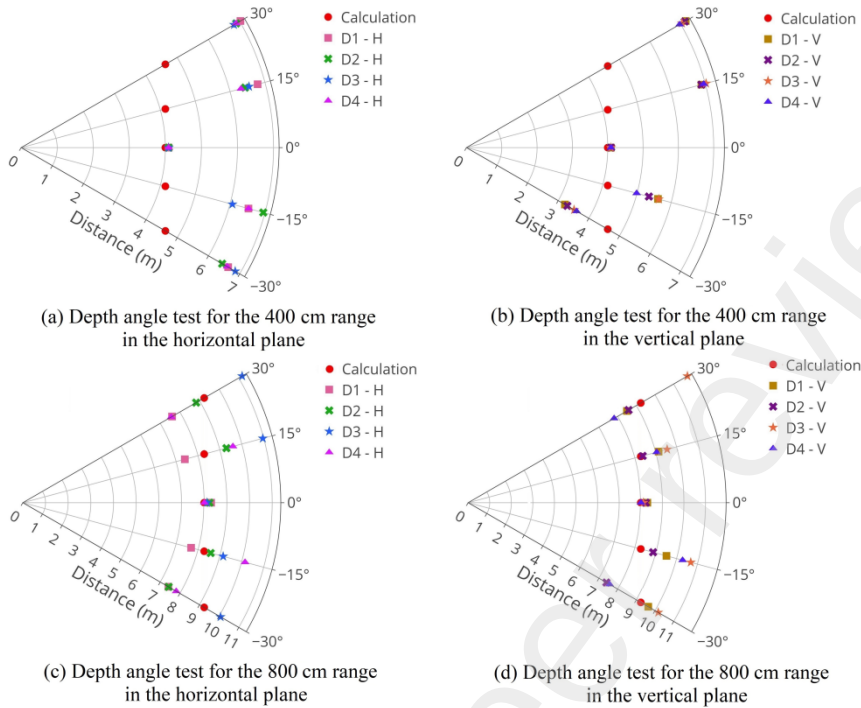


Fig. 9. The results of target detection in certain angle and distance

The red dot shows the calculation or proposed distance results and is compared with the measurement results through a depth-sensing camera. The result shows a measurement accuracy of up to 98% if the target is in line with the camera position, whereas if there is a difference in angle, there is a sizeable error of 15% – 45%. The difference in X_d distance produces different values, so the programming algorithm cannot resolve this reading error. Instead, it can be overcome by integrating a depth-sensing camera with a tracker and laser.

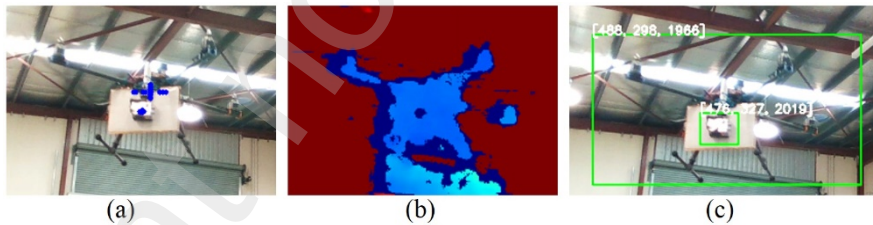


Fig. 10. Drone detection test: (a) Depth-sensing camera point (b) Depth Image (c) Coordinate axes.

We conducted experiments to achieve localization on three coordinate axes. The target is viewed at multiple points with a depth-sensing camera, and the distance is measured from this point to the depth-sensing camera sensor. Meanwhile, the target position in the bounding box is translated based on the centre of the bounding box. This central point compensation provides the target position in horizontal (Y-axis) and vertical axis (Z-axis) in pixel units, as shown in Fig. 10. The drone bounding box information is shown as $Y = 488$ px, $Z = 298$ px, and $X = 1966$ cm. This also applies to the drone camera bounding boxes. When tracking the drone camera, the accuracy of the readings is increased by always following the central point of the sensor.

4.4. Camera detection-based algorithm

The previous results influence the adaptive tracker's design, where two methods can be used to detect the target. The first method is based on the distance. When the drone distance is less than 500 cm, the position of the drone camera is detected through the position data from the depth-sensing camera, and laser irradiation is carried out according to the coordinates given. If the drone is farther than 500 cm, the drone camera cannot be detected. Hence, the intended target is the drone itself, and the laser irradiation is carried out 3 cm below the center of the bounding box (Z-axis). This value is selected based on data from commercial drones with cameras; generally, the camera position is 0.5 – 4 cm below the center heading of the drone.

Since the 500 cm threshold is dependent on the depth sensing camera that we tested, then a more hardware independent second method is through bounding box detection. If the system detects the drone camera with confidence greater than 0.7, the drone camera coordinate information is taken as input for the tracker to move. And, if only the drone is detected by the system, the coordinate of the drone is used to predict the position of camera using Equation 6. Testing this method is left as future work.

5. Laser Dazzling System Analysis

5.1. Tracking accuracy of laser dazzling system

A laser pointer is attached to the tracker to determine the laser's position on the camera. Laser pointers have a more clearly visible beam, to allow the laser's position to be visually tracked when following the drone camera and for the error of the laser pointing to be measured. The experiments to follow the drone camera's position are pictured in Fig. 11. The whiteboard matrix is used to verify the laser's position against the drone camera. The laser pointer's progress to track the drone's movement can be seen in Fig. 11(a). While the drone is moving, the laser pointer cannot remain at the center of the camera view due to a data processing delay in the time to read the drone's position. Fast target-searching patterns are not the focus of this research because a motor tracker with a higher rate is needed to follow high-speed drone motion, using any pattern of search targets. As shown in Fig. 11(b), the drone camera lens used in this study is 3.8 cm in diameter, where the horizontal axis is symbolized by Y, and the vertical axis is represented by Z. This experiment aims to determine the magnitude of the error between the laser position and the center field of view of the depth-sensing camera. When the drone moves, the tracker will dynamically follow the movement and try to place the target at the midpoint of the field of view.

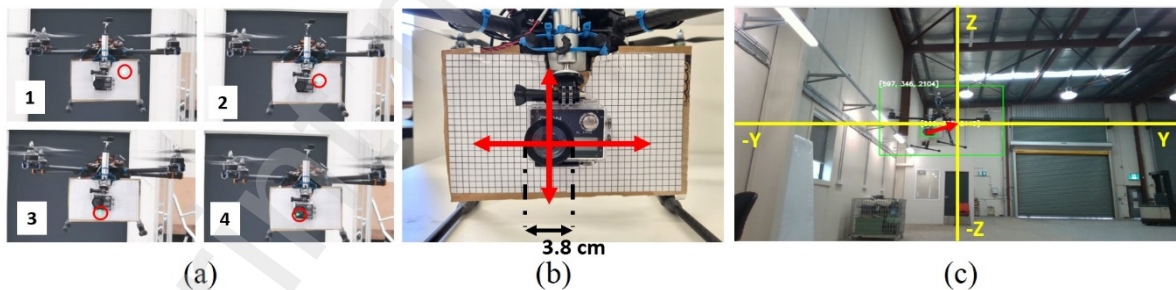


Fig. 11. Drone camera tracker experiment: (a) Laser position (b) Matrix board (c) Depth-sensing camera field of view.

When the drone is stationary, the laser can dazzle the drone camera effectively. The adaptive target tracker is designed to detect drone cameras when the sensing distance is less than 500 cm, and if it is greater than 500 cm, then the drone itself is tracked (see Fig. 8). Both operation modes are expected to be able to follow the movement

of the drone, and the laser point must remain near the lens or aimed at the camera. The trials were carried out with drones flying between 2 m/s to 4 m/s horizontally and vertically. The field of view of the depth-sensing camera is only 60° at about 400 cm. Thus, if the drone moves at high speed, it rapidly moves out of sight, which causes the tracker to return to searching mode instead of tracking. When the drone moves, it takes time for the tracker to reposition it at the midpoint of the depth-sensing camera field of view.

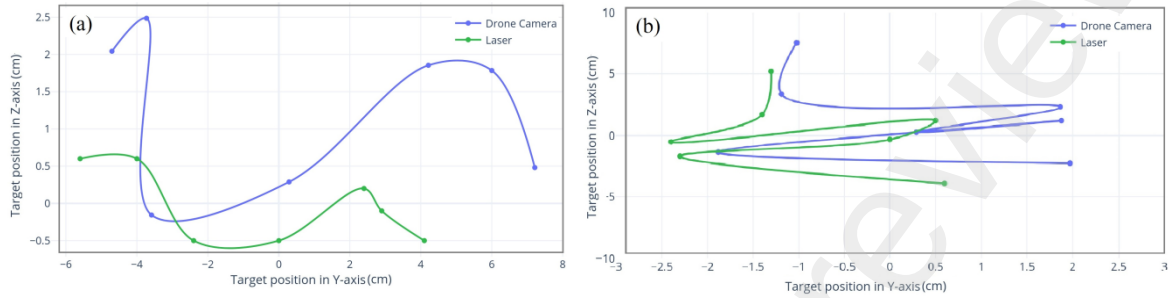


Fig. 12. The tracker's ability result to track the drone camera at a 400 cm distance in (a) The horizontal plane (b) The vertical plane.

Fig. 12(a) shows the results of horizontal testing, when the drone moves from left to right at a distance of 400 cm from the depth-sensing camera. From the matrix board, the camera's position and laser position in the (Y, Z) plane can be tracked. The depth-sensing camera generates position data in pixels that are converted into cm. The laser follows the drone camera with a maximum error of 3 cm. The 2-axis tracker results show that the average error on the Y-axis is greater than on the Z-axis in the horizontal plane. At short range, the depth-sensing camera can detect the drone camera and follow the drone movement at a low speed. The non-linear results on the camera position input are caused by the vertical or horizontal direction of the drone, which leads to a slight shift of its heading. The second experiment is to track the drone when it moves vertically with a speed between 2 m/s to 4 m/s, with a sensing distance of 400 cm, and the results are shown in Fig. 12(b). The tracker can continuously follow the drone's movements with a position difference of between 0.5 cm to 3.5 cm between the drone camera and the laser. As the drone moves, the tracker must reposition. We find the magnitude of the error on the Y-axis is greater than that on the Z-axis, with an average error of 1.95 cm, where the scattering laser can dazzle the camera even though the laser is not at the center of the lens (see Fig. 14). This can happen if the laser divergence is sufficient for a given range.

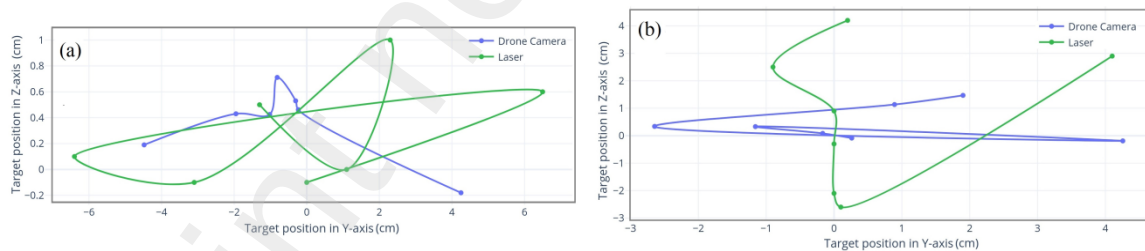


Fig. 13. The tracker's ability result to track the drone camera at a 1000 cm distance in (a) The horizontal plane (b) The vertical plane.

At long range, the trials are performed at 1000 cm, and the drone moves continuously in the horizontal and vertical planes. The laser set up is established in the drone's static mode before continuous tracking is carried out. If a drone is detected, the system receives the drone coordinates (Y, Z). Using Equation 6, the laser's position can be adjusted, and the location of the drone camera can be predicted. Unlike the results shown at

short range, the tracking error for the long-range trials is more significant, as shown in Fig. 13. The camera and laser positions should be at the exact coordinates, but there is a range difference of 1.7 cm to 6.2 cm. The observed size of the drone camera at long range is small and continuous tracking is not possible at high speed. Therefore, the test was carried out for low drone speed of 2 m/s. We found the maximum range between the drone camera and laser was 6.2 cm.

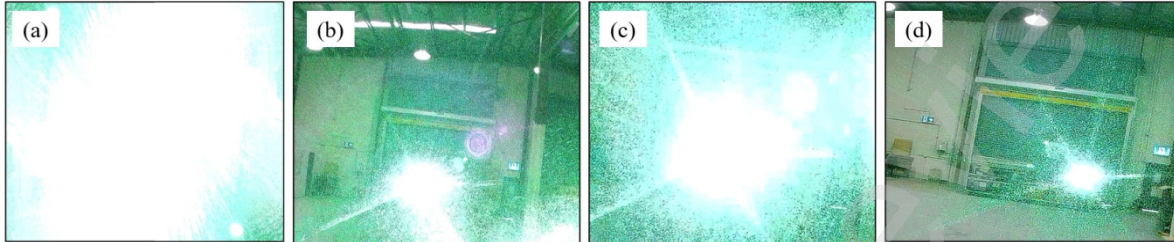


Fig. 14. The sample of dazzle image using a laser pointer in (a) 4 m range and within the camera lens field (b) 4 m range and out of the camera lens field (c) 10 m range and within the camera lens field (d) 10 m range and out of the camera lens field.

The effect of laser irradiation using a laser pointer can be seen in Fig. 14. In the irradiation on the drone camera for short or long range; the image can still be dazzled even though the point of the laser does not hit the center of the camera, provided that the laser hit a part of the lens. For short range, if the laser deviates up to 3 cm from the camera's center point, the image can still be dazzled, and the direction of the laser can still be seen. Meanwhile, if it deviates more than 3 cm or out of the camera lens for a long range, then only the light from the laser is visible while the image can still be seen.

5.2. Laser dazzling effect and incident angle

In this section, we study the influence of various factors on the effectiveness of dazzling, such as laser power and the incident angle of the laser on the CMOS camera sensor. In order to able to adjust the power, this experiment was performed using a green laser with a 520 nm wavelength to determine the system's ability to dazzle the CMOS camera. The camera has a total number of pixels of 4072 (Horizontal) x 3176 (Vertical) or approximately 12.93 MP, with a 6 mm focal length lens and infra-red filter. The test was carried out in an indoor lab with the camera positioned at a distance of 2 m in line with the laser. We observed light diffraction and scattering on the camera with background illumination (Bi) and ambient light ($Bi = 180$ lux) and low light ($Bi < 30$ lux). The results can be seen in Fig. 15. The laser output power (P) is set to specified levels. Under ambient light, dazzling and scattering appear on the camera and can dazzle the camera at 15.4 mW. However, in dark conditions, the pattern of laser scattering is evident at 7.1 mW. The power of 15.4 mW was selected and tested with a QR code background to optimize the system in dazzling the camera.

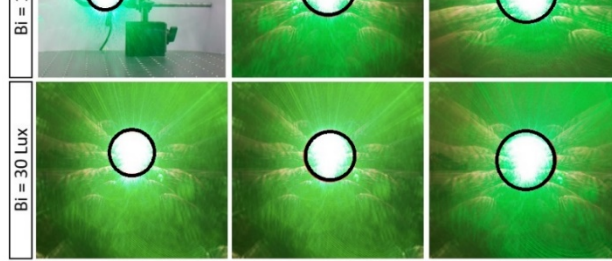


Fig. 15. Dazze image obtained using a green laser in different light condition.

The drone camera was dazzled to test when a QR code was no longer readable by software using the camera. The effect of the laser irradiance and power on the dazzled QR code captured by the CMOS camera in ambient light is presented in Fig. 16. By increasing the laser power from 7.1 mW to 15.4 mW, the radius of the dazzling point increases from 0.5 image scale to 4 image scale (or eight times stronger glare). Scattering occurs, which makes the background challenging to see clearly.

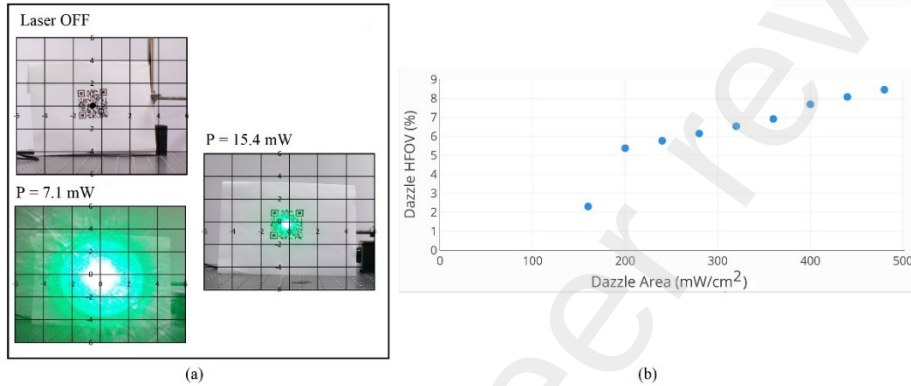


Fig. 16. Dazzled test: (a) Dazzled QR code (b) Dazze spot.

The saturation of the image is quantified via the dazzled camera area which is quantified using the percentage of horizontal field of view (HFOV) (Nascimento Santos et al. 2018a). The curve shows an increase in dazzled area for fluences up 0.1 mW/cm², but the increase saturates for higher irradiances as noted previously in (Nascimento Santos et al. 2018a). The effect of laser incidence angle was tested using the DJI mini at 500 cm. The camera is rotated clockwise to get the maximum rotation angle where it can still be dazzled by the laser. The trial layout is illustrated in Fig. 5, and the test results are shown in Fig. 17 and Fig. 18. With a laser power of 23.5 mW, the laser scattering causes overexposure at the camera at angles of 30° and 60°.

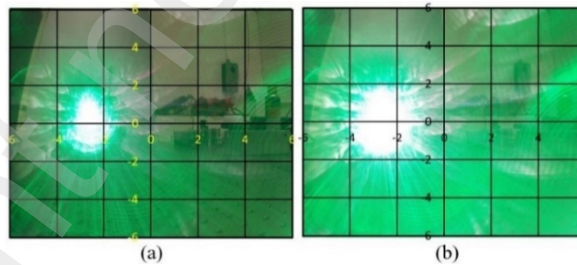


Fig. 17. Dazze images as a function of the laser incidence angle: (a) 30° with 15.4 mW (b) 30° with 23.5 mW.

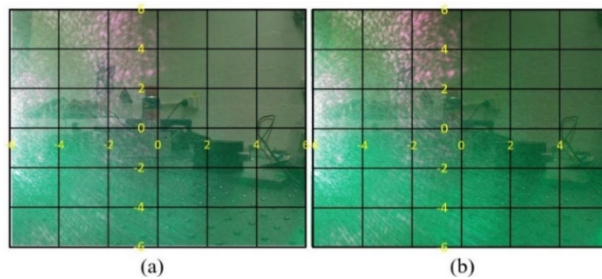


Fig. 18. Dazzle images as a function of the laser incidence angle: (a) 60° with 15.4 mW (b) 60° with 23.5 mW.

Effective laser-induced dazzling can be achieved with a suitable combination of the laser power and incidence angle. When applied to drones, continuous dazzling can be done up to a 60° angle of incidence of the laser. Thus, the laser plane and camera plane may not be parallel, but the laser beam should shine directly onto the camera lens. The laser beam divergence is calculated using Equation 12. At a power of 23.5 mW and a camera rotation of 30° the beam radius was 1.2 cm, leading to a beam divergence of 1°.

6. Conclusion and future direction

The development of two rotational axis trackers based on computer vision can detect the drone's appearance, localize the drone's position and precisely detect the camera's position on the drone. This makes it easier for the laser to dazzle the drone camera. The 520 nm green laser module wavelength with 23.5 mW power can disrupt drone vision. To dazzle the camera, the laser does not have to be directed straight to the drone camera but could have an angle of incidence of up to 60°. The scattered beam can dazzle the camera, provided it is still in line with the laser. Our study overcomes the challenge of detecting small objects, i.e., drone cameras, and identifying them with high accuracy from a long distance. The sensing device is a 435i depth-sensing camera based on the YOLOv5 object detection algorithm, which can detect drone cameras up to 500 cm with a confidence value of more than 0.7. These results were obtained using only 4000 drone camera datasets for training and testing. The confidence score can be further enhanced by adding training data to the augmentation data. Overall, object detection performance for drones and drone cameras reaches almost 0.9 accurate positive detections with a confidence loss level of less than 0.01. On the other hand, the device can sense the presence of a drone up to 2100 cm for a drone with a size of 69 cm. At a range of 800 cm, all types of drones with a wheelbase of 23 - 69 cm can always be detected. On the other hand, the position reading accuracy is 98% when the target is at the center of the field of view (FoV) of the depth-sensing camera. If the target is on the edge of the FoV, the error can be up to 45%. The adaptive detection method is used to detect both drones and drone cameras. If the drone camera is detected, the coordinates of the drone camera are used to aim the laser. If only drones are detected, then the drone's position is used to predict the drone camera's position. In the system verification process, the laser can hit the target perfectly when the drone hovers. But the focus of this research is to follow the movement of the drone continuously, and the result is that for a range of less than 5 m, the laser position is not always at the center of the drone camera but shifts up to 1.9 cm to follow the drone at low speed. High-power lasers have a large beam size, so this shift can be overcome if there is reflection from the drone body. In long-range detection, the resulting error is up to 6.2 cm. This is because the diameter of the drone camera is small, and it is challenging to detect even moving at low speed.

For further developments, we want to improve the vision sensors to increase the detection distance, and to improve the machine learning with the YOLOV5 algorithm, by adding filters such as the Kalman filter to reduce false positives and increase the confidence level of detection. The NVIDIA Jetson microprocessor can speed up detection processing time, and the tracker can use a high-precision motor with a rotational movement of less than 1 degree. We will also explore tracking at higher drone mobility rates. The laser beam divergence can be manipulated to increase the beam distance without increasing power. We also want to investigate under what conditions to transition from drone camera tracking to drone body tracking by using the hardware independent bounding box method. Exploring a general search method for scanning the skies is also a direction for future work.

CRedit authorship contribution statement

Endrowednes Kuantama: Conceptualization, Formal analysis, and Writing – original draft. **Yihao Zhang:** Software and Data curation. **Faiyaz Rahman:** Investigation and Analysis. **Richard Han:** Conceptualization, Methodology, Supervision. **Judith M Dawes:** Supervision, Review, and Editing. **Rich Mildren:** Methodology and Conceptualization. **Phuc Nguyen:** Validation, Methodology. **Tasnim Azad Abir:** Writing and Analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- Ahmed F, Mohanta JC, Keshari A, Yadav PS (2022) Recent Advances in Unmanned Aerial Vehicles: A Review. *Arab J Sci Eng* 47:7963–7984. <https://doi.org/10.1007/s13369-022-06738-0>
- Ali A, Barnett L, Sparks D, et al (2019) COUNTER-DRONE SYSTEMS
- Alsanad HR, Sadik AZ, Ucan ON, et al (2022) YOLO-V3 based real-time drone detection algorithm. *Multimed Tools Appl* 81:26185–26198. <https://doi.org/10.1007/s11042-022-12939-4>
- Altawy R, Youssef AM (2017) Security, privacy, and safety aspects of civilian drones: A survey. *ACM Transactions on Cyber-Physical Systems* 1:. <https://doi.org/10.1145/3001836>
- Benoist KW, Schleijpen RHMA (2014a) Modeling of the over-exposed pixel area of CCD cameras caused by laser dazzling. In: *Technologies for Optical Countermeasures XI; and High-Power Lasers 2014: Technology and Systems*. SPIE, p 92510H
- Benoist KW, Schleijpen RHMA (2014b) Modeling of the over-exposed pixel area of CCD cameras caused by laser dazzling. In: *Technologies for Optical Countermeasures XI; and High-Power Lasers 2014: Technology and Systems*. SPIE, p 92510H
- Castrillo VU, Manco A, Pascarella D, Gigante G (2022) A Review of Counter-UAS Technologies for Cooperative Defensive Teams of Drones. *Drones* 6
- Cazzato D, Cimarelli C, Sanchez-Lopez JL, et al (2020) A survey of computer vision methods for 2d object detection from unmanned aerial vehicles. *J Imaging* 6
- Chamola V, Kotes P, Agarwal A, et al (2021) A Comprehensive Review of Unmanned Aerial Vehicle Attacks and Neutralization Techniques. *Ad Hoc Networks* 111
- Chen H, Zhao X, Tan M (2015) A novel pan-tilt camera control approach for visual tracking. In: *Proceedings of the World Congress on Intelligent Control and Automation (WCICA)*. Institute of Electrical and

- Electronics Engineers Inc., pp 2860–2865
- Eberle B, Kinerk WT, Koerber M, et al (2019) NATO SET-249 joint measurement campaign on laser dazzle effects in airborne scenarios. *SPIE-Intl Soc Optical Eng*, p 11
- He K, Gkioxari G, Dollar P, Girshick R (2017) Mask R-CNN. In: *Proceedings of the IEEE International Conference on Computer Vision*. Institute of Electrical and Electronics Engineers Inc., pp 2980–2988
- Horaud R, Hansard M, Evangelidis G, Menier C (2020) An Overview of Depth Cameras and Range Scanners Based on Time-of-Flight Technologies. <https://doi.org/10.1007/s00138-016-0784-4>
- Hossain S, Lee DJ (2019) Deep learning-based real-time multiple-object detection and tracking from aerial imagery via a flying robot with GPU-based embedded devices. *Sensors (Switzerland)* 19:. <https://doi.org/10.3390/s19153371>
- Huang H, Gartner G (2018) Current trends and challenges in location-based services. *ISPRS Int J Geoinf* 7
- Huynh T, Tran MT, Lee DH, et al (2021) A study on vision-based backstepping control for a target tracking system. *Actuators* 10:. <https://doi.org/10.3390/act10050105>
- Jeon GH, Lee JH, Sung YS, et al (2022) Cooperative Friendly Jamming Techniques for Drone-Based Mobile Secure Zone. *Sensors* 22:. <https://doi.org/10.3390/s22030865>
- Kapustina L, Izakova N, Makovkina E, Khmelkov M (2021) The global drone market: main development trends. *SHS Web of Conferences* 129:11004. <https://doi.org/10.1051/shsconf/202112911004>
- Lin C, Zhang W, Shi J (2021) Tracking Strategy of Unmanned Aerial Vehicle for Tracking Moving Target. *Int J Control Autom Syst* 19:2183–2194. <https://doi.org/10.1007/s12555-020-2049-4>
- Liu H, Fan K, Ouyang Q, Li N (2021) Real-time small drones detection based on pruned yolov4. *Sensors* 21:. <https://doi.org/10.3390/s21103374>
- Lykou G, Moustakas D, Gritzalis D (2020) Defending airports from uas: A survey on cyber- attacks and counter-drone sensing technologies. *Sensors (Switzerland)* 20:1–35
- Madasamy K, Shanmuganathan V, Kandasamy V, et al (2021) OSDDY: embedded system-based object surveillance detection system with small drone using deep YOLO. *EURASIP J Image Video Process* 2021:. <https://doi.org/10.1186/s13640-021-00559-1>
- Maghazei O, Lewis MA, Netland TH (2022) Emerging technologies and the use case: A multi-year study of drone adoption. *Journal of Operations Management* 68:560–591. <https://doi.org/10.1002/joom.1196>
- Maru MB, Lee D, Tola KD, Park S (2021) Comparison of depth camera and terrestrial laser scanner in monitoring structural deflections. *Sensors (Switzerland)* 21:1–17. <https://doi.org/10.3390/s21010201>
- Merkert R, Bushell J (2020) Managing the drone revolution: A systematic literature review into the current use of airborne drones and future strategic directions for their effective control. *J Air Transp Manag* 89:. <https://doi.org/10.1016/j.jairtraman.2020.101929>
- Miikki K, Karakoç A, Rafiee M, et al (2021) An open-source camera system for experimental measurements. *SoftwareX* 14:. <https://doi.org/10.1016/j.softx.2021.100688>
- Mohsan SAH, Khan MA, Noor F, et al (2022) Towards the Unmanned Aerial Vehicles (UAVs): A Comprehensive Review. *Drones* 6
- Nascimento Santos C, Chretien S, Merella L, Vandewal M (2018a) Visible and near-infrared laser dazzling of CCD and CMOS cameras. *SPIE-Intl Soc Optical Eng*, p 27
- Nascimento Santos C, Chretien S, Merella L, Vandewal M (2018b) Visible and near-infrared laser dazzling of CCD and CMOS cameras. *SPIE-Intl Soc Optical Eng*, p 27
- Nguyen P, Truong H, Ravindranathan M, et al (2017) Matthan: Drone presence detection by identifying physical signatures in the drone's RF communication. In: *MobiSys 2017 - Proceedings of the 15th Annual International Conference on Mobile Systems, Applications, and Services*. Association for Computing Machinery, Inc, pp 211–224
- Paredes JA, Álvarez FJ, Aguilera T, Aranda FJ (2020) Precise drone location and tracking by adaptive matched filtering from a top-view ToF camera. *Expert Syst Appl* 141:. <https://doi.org/10.1016/j.eswa.2019.112989>
- Park HC, Chakir S, Kim YB, Huynh T (2021a) A nonlinear backstepping controller design for high-precision

- tracking applications with input-delay gimbal systems. *J Mar Sci Eng* 9:.
<https://doi.org/10.3390/jmse9050530>
- Park S, Kim HT, Lee S, et al (2021b) Survey on Anti-Drone Systems: Components, Designs, and Challenges. *IEEE Access* 9:42635–42659. <https://doi.org/10.1109/ACCESS.2021.3065926>
- Qadir A, Semke W, Neubert J (2014) Vision based neuro-fuzzy controller for a two axes gimbal system with small uav. *Journal of Intelligent and Robotic Systems: Theory and Applications* 74:1029–1047. <https://doi.org/10.1007/s10846-013-9865-z>
- Rao B, Gopi AG, Maione R (2016) The societal impact of commercial drones. *Technol Soc* 45:83–90. <https://doi.org/10.1016/j.techsoc.2016.02.009>
- Redmon J, Divvala S, Girshick R, Farhadi A (2016) You Only Look Once: Unified, Real-Time Object Detection
- Ren S, He K, Girshick R, Sun J (2015) Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks
- Ritt G, Eberle B (2015) Automatic laser glare suppression in electro-optical sensors. *Sensors (Switzerland)* 15:792–802. <https://doi.org/10.3390/s150100792>
- Rodríguez-Quinones JC, Sergiyenko O, Flores-Fuentes W, et al (2017) Improve a 3D distance measurement accuracy in stereo vision systems using optimization methods' approach. *Opto-Electronics Review* 25:24–32. <https://doi.org/10.1016/j.opelre.2017.03.001>
- Rohling H (Hermann), Deutsche Gesellschaft für Ortung und Navigation, Institute of Electrical and Electronics Engineers., Cuvillier Verlag (2017) MIMO OFDM Radar System for Drone Detection
- Rothe J, Strohmeier M, Montenegro S (2019) A concept for catching drones with a net carried by cooperative UAVs; A concept for catching drones with a net carried by cooperative UAVs
- Seidaliyeva U, Akhmetov D, Ilipbayeva L, Matson ET (2020) Real-time and accurate drone detection in a video with a static background. *Sensors (Switzerland)* 20:1–18. <https://doi.org/10.3390/s20143856>
- Serafinelli E (2022) Imagining the social future of drones. *Convergence* 28:1376–1391. <https://doi.org/10.1177/13548565211054904>
- Sheu BH, Chiu CC, Lu WT, et al (2019) Development of UAV tracing and coordinate detection method using a dual-axis rotary platform for an anti-UAV system. *Applied Sciences (Switzerland)* 9:. <https://doi.org/10.3390/app9132583>
- Souli N, Kolios P, Ellinas G (2022) An Autonomous Drone System with Jamming and Relative Positioning Capabilities. <https://doi.org/10.1109/ICC45855.2022.9838783>
- Svanström F, Alonso-Fernandez F, Englund C (2022a) Drone Detection and Tracking in Real-Time by Fusion of Different Sensing Modalities. *Drones* 6:317. <https://doi.org/10.3390/drones6110317>
- Svanström F, Alonso-Fernandez F, Englund C (2022b) Drone Detection and Tracking in Real-Time by Fusion of Different Sensing Modalities. *Drones* 6:317. <https://doi.org/10.3390/drones6110317>
- Tadic V, Toth A, Vizvari Z, et al (2022) Perspectives of RealSense and ZED Depth Sensors for Robotic Vision Applications. *Machines* 10
- Taha B, Shoufan A (2019) Machine Learning-Based Drone Detection and Classification: State-of-the-Art in Research. *IEEE Access* 7:138669–138682. <https://doi.org/10.1109/ACCESS.2019.2942944>
- Tiwari R, Dubey AK (2022) Detection of Camouflaged Drones using Computer Vision and Deep Learning Techniques. In: *Proceedings of the Confluence 2022 - 12th International Conference on Cloud Computing, Data Science and Engineering*. Institute of Electrical and Electronics Engineers Inc., pp 380–383
- Unlu E, Zenou E, Rivière N (2017) Using Shape Descriptors for UAV Detection
- Wang Y, Xia H, Yao Y, Huang Y (2016) Flying Eyes and Hidden Controllers: A Qualitative Study of People's Privacy Perceptions of Civilian Drones in The US. *Proceedings on Privacy Enhancing Technologies* 2016:172–190. <https://doi.org/10.1515/popets-2016-0022>
- Wang Z, Qi L, Tie Y, et al (2019) Drone Detection Based on FD-HOG Descriptor. In: *Proceedings - 2018 International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery, CyberC*

2018. Institute of Electrical and Electronics Engineers Inc., pp 433–436
- Wisniewski M, Rana ZA, Petrunin I (2022) Drone Model Classification Using Convolutional Neural Network Trained on Synthetic Data. <https://doi.org/10.3390/jimaging8080218>
- Wyder PM, Chen YS, Lasrado AJ, et al (2019) Autonomous drone hunter operating by deep learning and all-onboard computations in GPS-denied environments. <https://doi.org/10.1371/journal.pone.0225092>
- Zhang Z, Zhang J, Shao B, et al (2015) Typical effects of laser dazzling CCD camera. In: Third International Symposium on Laser Interaction with Matter. SPIE, p 95430A
- Zhao J, Zhang J, Li D, Wang D (2022) Vision-based Anti-UAV Detection and Tracking
- Luo C, et al (2019) Numerical simulation of dazzling effects on CCD induced by a 532nm pulsed laser. SPIE-Intl Soc Optical Eng, p 102